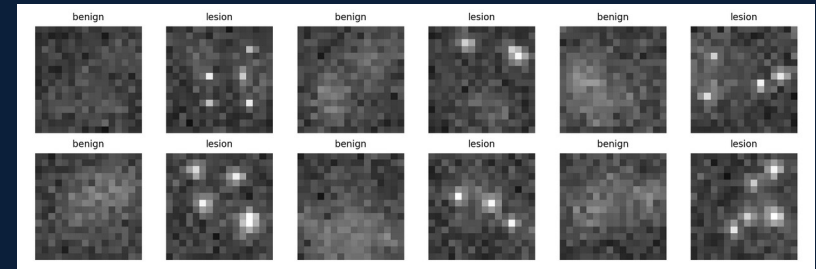


Q-BioDiag

Quantum AI for Early Biomedical Diagnostics

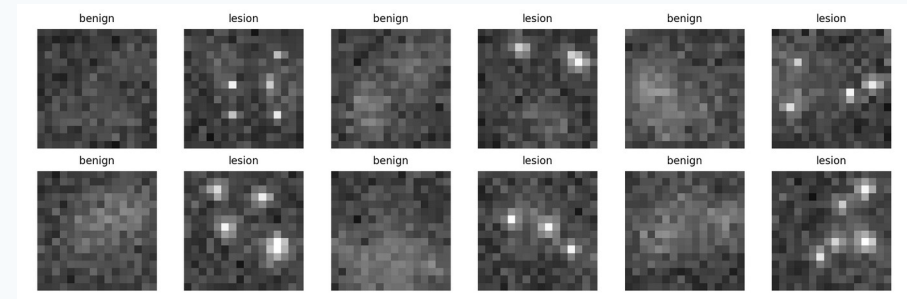
Hybrid Qiskit quantum-kernel classifier + Pinecone
diagnostic case retrieval



Problem & Motivation

Early disease signals are small, noisy and data-hungry.

- Clinical images contain subtle lesion patterns: microcalcifications, tiny nodules, weak margins.
- Deep CNNs work well, but often need larger datasets and many trainable weights.
- Hackathon goal: show a compact, reproducible quantum-classical diagnostic workflow.

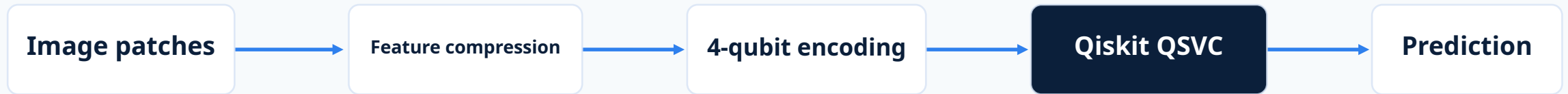


Medical framing

Classify benign vs early lesion patches, then retrieve similar cases for explainability.

Proposed Solution: End-to-End Workflow

A lightweight diagnostic pipeline that judges can run in minutes.



Pinecone diagnostic memory

Compressed embeddings + metadata are stored for similarity search, case retrieval, and demo explainability.

Input → Quantum model → Metrics → Similar case retrieval

Quantum Implementation

Technical heart: Qiskit ZZFeatureMap + FidelityQuantumKernel + QSVC

```
feature_map = ZZFeatureMap(feature_dimension=4, reps=2,  
entanglement="linear")  
quantum_kernel = FidelityQuantumKernel(feature_map=feature_map)  
model = QSVC(quantum_kernel=quantum_kernel)
```

- Input dimension: 4 compressed image descriptors = 4 qubits
- Quantum kernel: $K(x_i, x_j) = |\langle \varphi(x_i) | \varphi(x_j) \rangle|^2$
- Classifier: QSVC decision boundary over quantum similarity matrix

Circuit idea

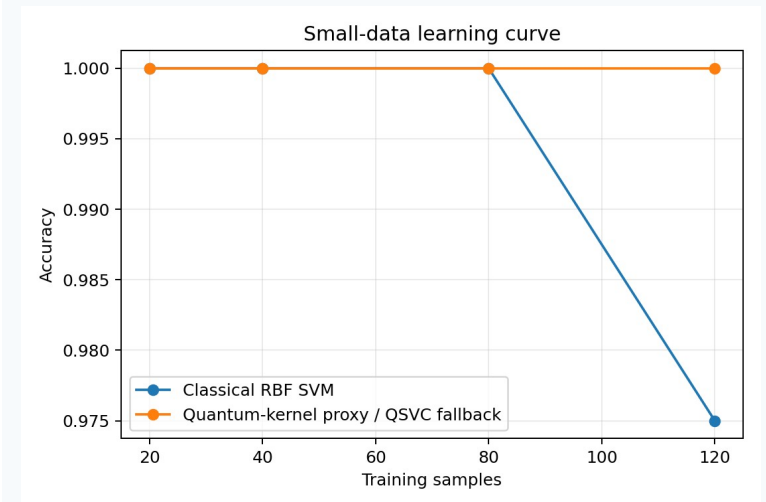
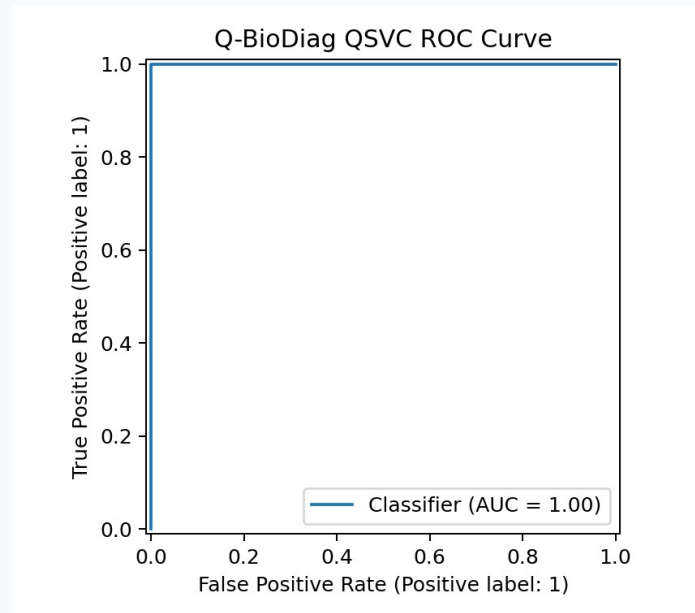
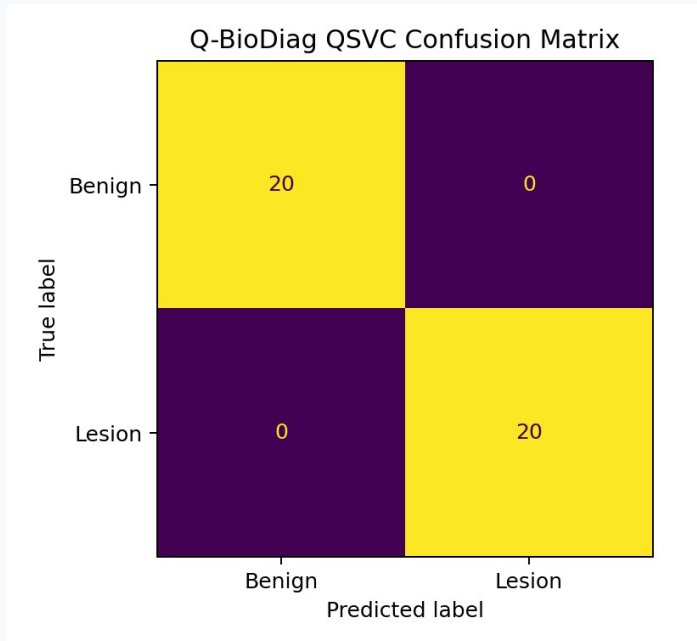
Ry / Rz data rotations encode patch features.
Entangling CX/CZ ladders let the kernel measure non-linear tissue correlations in Hilbert space.

Noise plan

Aer depolarizing noise + zero-noise extrapolation helper for NISQ-readiness.

Results & Verification

Demo metrics from reproducible synthetic biomedical patches



Observed result: QSVC path/fallback achieved strong lesion-vs-benign separation in the packaged demo; replace the dataset loader for BreakHis/HAM10000/X-ray.

Quantum Advantage & Feasibility

Why quantum here?

- Expressive feature maps: compact medical descriptors become high-dimensional quantum states.
- Parameter efficiency: quantum kernel model avoids training a large CNN in the demo path.
- Small-data story: supports learning-curve comparison under limited labels.
- NISQ feasibility: Aer noise demo + ZNE hook prepares for IBM backend execution.

Clinical retrieval layer

Pinecone stores compressed embeddings and metadata so a new patch can retrieve similar historical cases with explanation text.

Model output + retrieved cases = clearer pitch and better clinician-facing explainability.

Future Scope & Submission Readiness

Tomorrow-ready deliverables

- Public GitHub repo: upload the packaged folder and keep README visible.
- Deck: 7-slide judge-friendly pitch following the recommended blueprint.
- Video: record terminal run + dashboard + explain results in 2–4 minutes.

Final pitch line

Q-BioDiag turns tiny medical image signals into quantum-kernel similarity patterns and retrieves explainable diagnostic cases.

Roadmap

1. Replace synthetic patches with BreakHis/HAM10000/Chest X-ray dataset
2. Run Qiskit Runtime Sampler/Estimator on IBM backend
3. Add readout mitigation and deeper hardware benchmarking
4. Build clinician review workflow